



The effect of using internet hospitals on the physician-patient relationship: Patient perspective

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ABSTRACT

Background: Internet hospitals are rapidly developing in China. Despite many studies regarding internet hospitals, there has been little further research that evaluates the impact of using internet hospitals on the physician-patient relationship during outpatient visits.

Methods: We developed a questionnaire based on the patient-doctor relationship questionnaire (PDRQ-9) to survey the physician-patient relationship. A sample of 505 patients who sought medical services from offline physical or internet hospitals was selected by convenience sampling. Multiple linear regression analysis was performed to determine whether the use of internet hospitals during outpatient visits was associated with the physician-patient relationship.

Results: Internet hospital users gave significantly lower scores than nonusers in total physician-patient relationship scores ($P = .01$) and in the 5 items of "My physician helps me" ($P < .001$), "I trust my physician" ($P = .001$), "My physician understands me" ($P = .002$), "My physician and I agree on the nature of my medical symptoms" ($P = .01$), and "I can talk to (communicate with) my physician freely" ($P = .005$). Multiple linear regression results showed that the use of internet hospitals during outpatient visits influenced the physician-patient relationship. And after controlling for other patient characteristics, the use of internet hospitals decreased the physician-patient relationship scores by 11.9%.

Conclusions: Our findings suggest that the current use of internet hospitals could not significantly enhance the physician-patient relationship during outpatient visits. Therefore, we should work on improving physicians' online communication skills and strengthening the level of trust between physicians and their patients. Policy-makers should also pay close attention to the gap of the physician-patient relationship between internet hospitals and offline physical hospitals.

1. Introduction

With an ever-increasing population suffering from chronic and age-related diseases, healthcare requirements worldwide have become a major challenge for governments, especially in China [1,2]. Internet hospitals, an innovative approach to health service delivery via internet technology [3-5], are emerging in China as a potential solution to optimize the allocation of medical resources and meet the increasing

public demands for high-quality medical services since October 2014 [3,5-7]. Subsequently, a series of support policies, promulgated by Chinese government from 2015 to 2019, greatly accelerated the development of internet hospitals [8-12]. In particular, internet hospitals played a significant role in preventing and controlling COVID-19 in China, which further triggered the growth of internet hospitals [13].

Internet hospitals refer to one-stop online service platforms that integrate online consultation, prescription, payment and drug delivery

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based on offline physical hospitals [3-5,7]. Patients in Internet hospitals are mainly those with mild and chronic illnesses who seek follow-up consultations, rather than those with acute and serious illnesses, and they would visit the corresponding offline physical hospitals for further medical examinations or treatments if necessary, thus potentially improving the patient's continuity of online to offline medical services [3-5]. Patients in the internet hospitals have to complete a registration process similar to the outpatient clinics of offline hospitals, such as filling out forms about their conditions, the course of their diseases and the medical treatment they seek, and uploading previous medical records, which help the physicians better to understand their conditions [14]. Internet hospitals overcome geographical obstacles and shatter time barriers, allowing patients to fully communicate with skilled experts via their computers or smartphone APPs consultation forms, such as graphic messages, phone calls or video chats. Therefore, compared to offline physical hospitals, internet hospitals can alleviate the difficulty of long waiting times for patients, increase the time that the physician spend with the patient and improve the accessibility of high-quality medical services. However, many existing problems need to be solved in internet hospitals, including surveillance and quality control, division of medical liability, information security and privacy protection [3,5,15].

Previous research has shown that internet hospitals, where medical experience is fairly comparable to that of offline physical hospitals [6,16], could adequately build the physician-patient relationship similar to that of offline physical hospitals [16]. Although medical workers made huge contributions to fighting against the COVID-19 epidemic, there were several incidents of violence against them and the physician-patient contradiction remains a prominent social issue in China. Hence, the assessment of the effect of internet hospitals on the physician-patient relationship is critical for promoting the development of internet hospitals and easing the current tense physician-patient relationship.

Existing research on internet hospitals mainly focused on their development status, service model, medical service capability and patient satisfaction [4,7,17-19], as well as their role in COVID-19 epidemic prevention and control [15,20,21]. Several studies have also identified the determinants of patients' intention to use internet hospitals [5] and different types of internet hospitals [8]. Some studies investigated patients' perceived barriers and facilitators to the adoption of internet hospitals [22], and analyzed the operations and functional positioning of developing internet hospital medical services in relation to physical hospitals [23].

Despite the rapid development and increasing social acceptance of internet hospitals in China [24], evidence to evaluate their effectiveness in improving health care delivery compared to offline physical hospitals is lacking to date. Some survey studies from the perspective of patients or physicians have assessed the effect of using online social media (such as Facebook, and Twitter) [25,26] or online health information on the physician-patient relationship [27-30], but there has been little further research that evaluates whether the use of internet hospitals can improve the physician-patient relationship from the patient experience perspective.

The physician-patient relationship is the core of the patient-centered care model and a positive physician-patient relationship could help patients to comply with medical prescriptions, especially for these patients with chronic diseases, thereby affecting patient health outcomes [31]. As an important component of health care quality evaluation, the physician-patient relationship plays a crucial role in enhanced patients' satisfaction with their medical care [32]. Superior physician-patient relationships may lead to better patient outcomes [33,34], such as trust, mutual understanding, adherence, increased access to patient-centered care, greater patient knowledge and shared understanding, and higher quality medical decisions, where better communication plays an important role [35].

Therefore, this study aimed to assess the capacity of using internet hospitals to enhance the physician-patient relationship and to compare

the physician-patient relationship between internet hospital users and offline physical hospital users during outpatient visits.

2. Methods

2.1. Questionnaire design

The Patient-Doctor Relationship Questionnaire (PDRQ-9) was the basis of our survey. The PDRQ-9 is a well-validated (Cronbach alpha = 0.921; $\chi^2/df=3.799$, RMR = 0.020, RMSEA = 0.092, GFI = 0.935, CFI = 0.958, TLI = 0.944 and NFI = 0.949 verified in this study), brief and widely-used tool for measuring the physician-patient relationship, comprising nine items regarding different aspects of physician-patient relationship (helpfulness, time sufficiency, trustworthiness, understanding, dedication and accessibility of the physician, how content the patient is with their care, and how much agreement exists about the nature of the patient's problems); and all items are rated on a scale of 1 (not at all) through 5 (totally), with a higher total score calculated by the sum of all item scores indicating a better physician-patient relationship [35]. We accordingly developed a survey questionnaire for measuring the physician-patient relationship in the users of internet hospitals and physical hospitals (nonusers of internet hospitals) during outpatient visits in China. All the items were translated from English to Chinese for the PDRQ-9 by an expert proficient in both Chinese and English, further discussed and approved by a focus group based on their expertise, and finally back-translated from Chinese to English by another qualified translator. The questionnaire items (Textbox 1) formed in the study were measured with a 5-point Likert scale ranging from "strongly disagree" (1) to "strongly agree" (5).

We performed a presurvey to validate the developed questionnaire in 67 patients (14 users of internet hospitals). Context-specific adjustments were then made according to the feedback from the pilot survey. Data on participant characteristics (see Table 1), as well as visit information (e. g., did you choose an internet hospital during this latest visit? —to divide the users and nonusers of internet hospitals) were also collected.

2.2. Data collection

The target population was adult patients who sought medical services from physical or internet hospitals, selected using the convenience sampling. Due to the influence of the ongoing COVID-19 epidemic, we further used the web-based survey tool called *wenjuanxing*, to create an electronic questionnaire in which to survey targeted patients [36]. From September 11 to October 11, 2021, we sent the survey link to Tencent QQ groups, WeChat groups, and Baidu Post Bar. Participants could click on the survey link using their phones to access and complete the questionnaire. Before the survey, we introduced the purpose of the survey, provided the definition of internet hospitals, and guaranteed that the survey data would not be used for other purposes. After an individual's consent was obtained, the survey was conducted accordingly.

2.3. Statistical analysis

Descriptive statistics were performed to summarize data on participant characteristics. T-tests and Mann-Whitney U tests were used to test the mean difference in perceived physician-patient relationship scores between users and nonusers of internet hospitals. Multiple linear regression analysis was further performed to determine whether the use of internet hospitals was associated with a better physician-patient relationship during the outpatient visit, and therein, the independent variables on the participant characteristics were set as control variables to reduce the effects of confounding factors.

Textbox 1

. Measurement items of the Chinese version of physician-patient relationship questionnaire in this study.

- Q1. My physician helps me
- Q2. My physician has enough time to provide the diagnosis or treatment for me
- Q3. I trust my physician
- Q4. My physician understands me
- Q5. My physician is dedicated to help me
- Q6. My physician and I agree on the nature of my medical symptoms
- Q7. I can talk to (communicate with) my physician freely
- Q8. I feel content with my physician's diagnosis or treatment
- Q9. I find my physician easily accessible

3. Results

3.1. Patient characteristics of internet hospital users and nonusers

In total, 543 responses were received, and 505 eligible responses were remained. Detailed characteristics of participants are shown in [Table 1](#). Among these internet hospital users, they accessed outpatient service via graphic message (87.7%, 114/130), telephone (7.7%, 10/130), or video chat (4.6%, 6/130) from the internet hospitals. The internet hospital users and nonusers differed in gender ($\chi^2 = 10.981$, $P = .001$), whereas there were no significant differences in other characteristics between the 2 groups.

3.2. Differences in physician-patient relationship between internet hospital users and nonusers

The analysis results showed that there was a significant difference in total physician-patient relationship scores, and the 5 items between the 2 groups ([Table 2](#)).

In total physician-patient relationship scores, internet hospital users gave significantly lower scores than nonusers ($P = .01$). In the items "My physician helps me" ($P < .001$), "I trust my physician" ($P = .001$), "My physician understands me" ($P = .002$), "My physician and I agree on the nature of my medical symptoms" ($P = .01$), and "I can talk to (communicate with) my physician freely" ($P = .005$), internet hospital users obtained significantly lower scores than nonusers as well, whereas other four items were all not significantly different between the 2 groups.

3.3. Influence of use of internet hospitals on physician-patient relationship

As shown in the [Table 3](#), whether internet hospital was used or not during outpatient visits was a significant predictor that influenced physician-patient relationship scores ($\beta = -1.391$, $P = .006$) after controlling for other factors on patient characteristics, and the standardized coefficient of whether internet hospital was used was -0.119 , indicating that when other covariates were held constant, the use of internet hospitals decreased the physician-patient relationship scores by 11.9%.

Age ($\beta = -0.102$, $P = .030$) and occupation ($\beta = -2.010$, $P = .007$; $\beta = -2.353$, $P = .025$, respectively) were all significant factors influencing the physician-patient relationship scores, whereas whether there was a chronic disease had no significant effect on the physician-patient relationship. Patients with higher age were more likely to have a lower physician-patient relationship score, and patients who were professional and technical staff or students were also more likely to have a lower physician-patient relationship score during the outpatient visit. Moreover, compared to those who rated their health status as "fair", patients who rated their health status as "good" or "very good" would be more

likely to have a higher physician-patient relationship score ($\beta = 1.987$, $P < .001$; $\beta = 4.781$, $P < .001$, respectively).

Moreover, all independent variables included in the model explained 20.54% of the variance in the physician-patient relationship score ($R^2 = 0.2054$). We also checked for collinearity by calculating the value of variance inflation factor, with a mean of 1.84, indicating there was no collinearity.

4. Discussion

We found that internet hospital users gave significantly lower physician-patient relationship scores than nonusers, indicating that the current use of internet hospitals could not significantly improve the physician-patient relationship during outpatient visits. In our study, internet hospital users had a lower experience score in the aspects of helpfulness, trust, understanding, agreement and communication, which may be relevant to the numerous existing problems to be solved in internet hospitals [3,5,15]. For example, an effective quality control system for internet hospitals has not been established in China, and a lack of regulations, supervision procedures, and methods for Internet diagnosis and treatment could increase the likelihood of medical and information risks, decrease patient trust and thus have a negative impact on risk-conscious patients' willingness of using online medical services [5,37]. Currently, the treatment behavior control in internet hospitals is still relatively loose, and neither the patient nor the physician can achieve the expected treatment effect via the internet [37]. Specifically, owing to the lack of face-to-face communication like the offline, patients felt they could not get the clear explanations and treatment as they did in offline physical hospitals, and physicians also thought they could not express themselves clearly to their patients, resulting in lower communication efficiency with certain patients who repeatedly asked them [37]. Such a change compared to offline physical hospitals might have a profound effect on the physician-patient relationship. Another possible reason could be relevant to the fact that physicians are increasingly being requested to consider emotional support while meeting their patients' communicative needs, whereas patients who used medical services from offline physical hospitals can receive more emotional support from their physicians during the face-to-face consultation compared to those from internet hospitals, thereby improving their satisfaction [38]. Moreover, previous research revealed that online health information seeking had a positive effect on the physician-patient relationship, when the patient could discuss the health information with the physician [30], and that good physician-patient communication can further lead to better patient health outcomes (e.g., trust, mutual understanding, adherence, and enhanced physician-patient relationship) [35]. Our findings might suggest that internet hospitals are still far behind physical hospitals including but not limited to the service content, quality

Table 1

Difference in participant characteristics of internet hospital users and nonusers (N = 505).

Characteristic	Overall, n (%)	Internet hospital users, n (%)	Nonusers, n (%)	χ^2 (df)/t-test	P ^a value
Gender				10.981 (1)	0.001
Male	214 (42.4)	39 (30.0)	175 (46.7)		
Female	291 (57.6)	91 (70.0)	200 (53.3)		
Age (Mean $\pm$ SD ^b)	28.24 $\pm$ 7.04	28.09 $\pm$ 5.94	28.29 $\pm$ 7.39	0.269	0.79
Educational level				2.882 (2)	0.24
High School and below	44 (8.7)	7 (5.4)	37 (9.9)		
College or undergraduate	259 (51.3)	66 (50.8)	193 (51.5)		
Postgraduate and above	202 (40.0)	57 (43.8)	145 (38.7)		
Occupation				4.996 (5)	0.42
Administrative staff	58 (11.5)	19 (14.6)	39 (10.4)		
Professional and technical staff	177 (35.0)	48 (36.9)	129 (34.4)		
Worker/service staff	36 (7.1)	8 (6.2)	28 (7.5)		
Students	151 (29.9)	33 (25.4)	118 (31.5)		
Unemployed/retired	28 (5.5)	10 (7.7)	18 (4.8)		
Others	55 (10.9)	12 (9.2)	43 (11.5)		
Monthly income, RMB (US \$)				0.740 (3)	0.86
$\leq$ 1999(309.29)	124 (24.6)	35 (26.9)	89 (23.7)		
2000 (309.44)-4999 (773.46)	123 (24.4)	29 (22.3)	94 (25.1)		
5000 (773.610-7999 (1237.62)	112 (22.2)	28 (21.5)	84 (22.4)		
$\geq$ 8000(1237.78)	146 (28.9)	38 (29.2)	108 (28.8)		
Marital status				3.145 (2)	0.08
Unmarried	344 (68.1)	91 (62.3)	263 (70.3)		
Married	158 (31.3)	49 (37.7)	109 (29.1)		
Others	3 (0.6)	—	3 (0.8)		
Current living place				0.000 (1)	0.98
Urban areas	447 (88.5)	115 (88.5)	332 (88.5)		
Rural areas	58 (11.5)	15 (11.5)	43 (11.5)		
Types of medical insurance				2.422 (3)	0.49
Basic medical insurance for urban employees	256 (50.7)	70 (53.8)	186 (49.6)		
Basic medical insurance for urban and rural residents	175 (34.7)	39 (30.0)	136 (36.3)		
Commercial medical insurance	23 (4.6)	8 (6.2)	15 (4.0)		
Others	51 (10.1)	13 (10.0)	38 (10.1)		
Self-rated health status				1.873 (3)	0.60
Poor	12 (2.4)	5 (3.8)	7 (1.9)		

Table 1 (continued)

Characteristic	Overall, n (%)	Internet hospital users, n (%)	Nonusers, n (%)	χ^2 (df)/t-test	P ^a value
Fair	144 (28.5)	35 (26.9)	109 (29.1)		
Good	257 (50.9)	65 (50.0)	192 (51.2)		
Very good	92 (18.2)	25 (19.2)	67 (17.9)		
Whether there was a chronic disease				2.180 (1)	0.14
Yes	36 (7.1)	13 (10.0)	23 (6.1)		
No	469 (92.9)	117 (90.0)	352 (93.9)		
Specialty services				9.118 (6)	0.17
Internal medicine	153 (30.3)	46 (35.4)	107 (28.5)		
Surgery	110 (21.8)	23 (17.7)	87 (23.2)		
Obstetrics and gynecology	49 (9.7)	9 (6.9)	40 (10.7)		
Dermatology	69 (13.7)	21 (16.2)	48 (12.8)		
Pediatrics	22 (4.4)	8 (6.2)	14 (3.7)		
Stomatology	41 (8.1)	6 (4.6)	35 (9.3)		
Others	61 (12.1)	17 (13.1)	44 (11.7)		
Physician title				0.599 (3)	0.90
Senior	112 (22.2)	27 (20.8)	85 (22.7)		
Deputy Senior	168 (33.3)	43 (33.1)	125 (33.3)		
Intermediate	174 (34.5)	48 (36.9)	126 (33.6)		
Junior	51 (10.1)	12 (9.2)	39 (10.4)		

^aP values of Chi-square tests for categorical variables, and t-test for continuous variables.^bSD: standard deviation.**Table 2**

Physician-patient relationship scores of internet hospital users and nonusers (n = 505).

Items	Internet hospital users scores (Mean $\pm$ SD ^a)	Nonusers scores (Mean $\pm$ SD)	P value ^b
Q1. My physician helps me	3.79 $\pm$ 0.67	4.07 $\pm$ 0.58	<0.001
Q2. My physician has enough time to provide the diagnosis or treatment for me	3.53 $\pm$ 0.86	3.54 $\pm$ 0.80	0.95
Q3. I trust my physician	3.73 $\pm$ 0.77	3.97 $\pm$ 0.62	0.001
Q4. My physician understands me	3.36 $\pm$ 0.87	3.63 $\pm$ 0.75	0.002
Q5. My physician is dedicated to help me	3.68 $\pm$ 0.78	3.81 $\pm$ 0.68	0.11
Q6. My physician and I agree on the nature of my medical symptoms	3.68 $\pm$ 0.74	3.86 $\pm$ 0.64	0.01
Q7. I can talk to (communicate with) my physician freely	3.66 $\pm$ 0.87	3.88 $\pm$ 0.68	0.005
Q8. I feel content with my physician's diagnosis or treatment	3.68 $\pm$ 0.86	3.81 $\pm$ 0.65	0.24
Q9. I find my physician easily accessible	3.59 $\pm$ 0.85	3.61 $\pm$ 0.78	0.85
Total perceived physician-patient relationship scores	32.71 $\pm$ 5.72	34.17 $\pm$ 4.85	0.01

^aSD: standard deviation.^bP values of Mann-Whitney U tests for continuous variables without a normal distribution, and t-tests for continuous variables with a normal distribution.

Table 3

Factors associated with physician-patient relationship scores in the multiple linear regression.

Variables	β	SE ^a	t test	P value
Constant	36.533	1.749	20.89	<0.001
Whether internet hospital was used	-1.391	0.508	-2.740	0.006
Gender (ref ^b : male)				
Female	-0.471	0.476	-0.99	0.323
Age	-0.102	0.0469	-2.18	0.030
Educational level (ref: postgraduate and above)				
High School and below	0.765	1.062	0.72	0.472
College or undergraduate	0.951	0.494	1.920	0.055
Occupation (ref: administrative staff)				
Professional and technical staff	-2.010	0.745	-2.70	0.007
Worker/service staff	-1.544	1.086	-1.42	0.156
Students	-2.353	1.049	-2.24	0.025
Unemployed/retired	-1.268	1.256	-1.01	0.313
Others	-1.480	0.981	-1.51	0.132
Monthly income (ref: $\leq$ 1999 RMB)				
2000–4999	0.876	0.730	1.200	0.230
5000–7999	-0.0836	0.878	-0.100	0.924
$\geq$ 8000	0.201	0.890	0.230	0.822
Marital status (ref: unmarried)				
Married	-0.168	0.679	-0.250	0.805
Others	N/A ^c	N/A	N/A	N/A
Current living place (ref: urban areas)				
Rural areas	0.755	0.735	1.030	0.305
Types of medical insurance (ref: basic medical insurance for urban employees)				
Basic medical insurance for urban and rural residents	0.543	0.610	0.890	0.374
Commercial medical insurance	-1.084	1.100	-0.990	0.325
Others	0.458	0.814	0.560	0.574
Self-rated health status (ref: fair)				
Poor	N/A	N/A	N/A	N/A
Good	1.987	0.515	3.86	<0.001
Very good	4.781	0.664	7.20	<0.001
Whether there was a chronic disease (ref: No)				
Yes	-0.627	0.891	-0.700	0.482
Specialty services (ref: internal medicine)				
Surgery	-0.702	0.613	-1.150	0.253
Obstetrics and gynecology	0.598	0.875	0.680	0.495
Dermatology	-0.295	0.712	-0.410	0.679
Pediatrics	-1.708	1.164	-1.470	0.143
Stomatology	-1.172	0.859	-1.360	0.173
Others	1.495	0.736	2.030	0.043
Physician title (ref: senior)				
Deputy Senior	-0.256	0.604	-0.420	0.672
Intermediate	-0.263	0.607	-0.430	0.664
Junior	-0.994	0.830	-1.200	0.231

^aSE: standard error.

^bRe: reference.

^cNA: not applicable.

and patient experience, and indicate that we should make full use of the advantages of Internet hospitals and narrow the gap to improve the physician-patient relationship.

Our findings contradicted results of several survey studies from the perspective of patients or physicians regarding the effect of online social media on the physician-patient relationship [25,26]. The possible reason for these differences might be explained partly by the different evaluated methods, where the effect of online social media on the physician-patient relationship was assessed based on a simple question that “can physician interaction with patient or patient relative through social media improve the doctor–patient relationship?” [25] or “how did social media influence the relationship with patients?” [26]. Previous research found that the physician-patient relationship could be improved through online social media or the internet by increasing the interaction between the physician and the patient [25,26,30], and online health information gathering also showed the potential to foster

greater patient engagement in health maintenance and care [29]. Patients from internet hospitals can easily find a targeted physician by searching for relevant information and have enough time to receive their physician’s diagnosis and treatment. However, communication problems between the physician and the patient can easily arise, because neither patients nor physicians from internet hospitals are familiar with each other. At the same time, the patient may worry that the physician would not have a comprehensive understanding of her/his medical symptoms like in offline physical hospitals, particularly during the first online visit, resulting in a lack of trust in her/his physician. Relevant research also showed that most physicians interacted with patients on social media outside working hours, and thus greatly improved their relationship with their patients [26], which puts forward higher requirements for physicians from internet hospitals to communicate with their patients.

Our study found that higher age was associated with a worse physician-patient relationship during the outpatient visit, which could be supported by the finding of the analysis of influencing factors of medical disputes that patients with higher age were more likely to experience medical disputes [39]. In addition, the older people prefer to see doctors they know and trust in traditional offline hospital outpatient clinics, since internet hospitals use new operation and service models, which are challenging for them. Our results also indicated that patients with worse self-rated health status were more likely to experience a negative physician-patient relationship, which is supported by the findings of the study by Shadmi et al. that older adults with high morbidity assigned a lower rating to physician-patient communication, knowledge of patient and trust in their physicians than those with moderate morbidity [40]. This finding might be explained partly by patients’ lack of trust in the online health services provided by internet hospitals. Previous research on the patients’ perceptions of barriers and facilitators to the adoption of internet hospitals indicated that older patients with long-standing diseases or poor health conditions were less likely to trust a new, less-mature health care delivery model, and these patients had a higher expectation for medical services and were more aware of the importance of regular thorough self-examinations and follow-ups [22]. Therefore, they may be less satisfied when they perceived the quality of medical services provided by internet hospitals was lower than that of offline physical hospitals, which could negatively affect the relationship with their physicians.

5. Limitations

There are several limitations mentioned in this study. First, data collection was self-reported by patients via an online survey due to the effect of the current ongoing COVID-19 epidemic, which might have a recall bias. Second, given that the internet penetration rate of the older people in China was still much lower than that of those younger people [41], possibly resulting in lower awareness of our online survey among those with older age, and the younger might be more willing to participate in the online survey [5], our conclusions might have been biased by the age distributions. After controlling for other patient characteristics, multiple linear regression analysis was used in this study to assess the impact of using the Internet hospital during outpatient visits on the physician-patient relationship, which could have resulted in stable and reliable conclusions. Third, some possible factors (e.g., wait times, whether seeing the same provider over time) were not considered as control variables in this study, which might affect the assessment of impact of using internet hospitals on the physician-patient relationship during outpatient visits to some extent. Furthermore, although our study analyzed the impact of using internet hospitals on the physician-patient relationship during outpatient visits by quantitative methods, further research is necessary to identify the root causes of the phenomenon and verify possible explanations through qualitative methods.

6. Conclusions

Our work provides evidence showing that the current use of internet hospitals could not significantly enhance the physician-patient relationship during outpatient visits. Therefore, we should work on improving physicians' online communication skills and strengthening the level of trust between them and their patients. Policymakers should also pay close attention to the physician-patient relationship gap between internet hospitals and offline physical hospitals, and devote to eliminating the negative effects of the current existing gaps with offline physical hospitals, including but not limited to service content, quality, and patient experience.

7. Summary table

1. What was already known on the topic

- Internet hospitals are rapidly developing in China, and play an increasingly important role in the delivery of health services, but evidence to evaluate their effectiveness in improving health care delivery compared to offline physical hospitals is lacking.
 - Physician-patient relationship is an important component of health care quality evaluation; however, the sharp physician-patient contradiction is still a prominent social problem in China.
 - Despite many studies regarding internet hospitals, there has been little further research that evaluates the effect of the use of internet hospitals on the physician-patient relationship during outpatient visits.
- ### 2. What this study added to our knowledge
- This study evaluates the capacity of the use of internet hospitals to enhance physician-patient relationship during outpatient visits from patients' perspective.
 - There remains empirical research on the effects of the currently existing gaps between internet hospitals and offline hospitals on the physician-patient relationship.

Ethics statement:

The study was approved by the Ethics Committee of Tongji Medical College of Huazhong University of Science & Technology (No. IORG0003571).

CRediT authorship contribution statement

Dehe Li: Formal analysis, Writing – original draft. **Yinhuan Hu:** Conceptualization, Funding acquisition, Project administration. **Sha Liu:** Methodology, Writing – review & editing. **Gang Li:** Resources, Validation. **Chuntao Lu:** Data collection. **Shaochun Yuan:** Data collection. **Zemiao Zhang:** Data collection. All authors approved the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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